fn score_candidates_map

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `score_candidates_map` in `mod.rs` at commit f5bb719 (outdated[1]). `score_candidates_map` returns a specific function that can be used to prove stability of the quantile scoring transformation.

1 Hoare Triple

Precondition

`alpha_den` > `alpha_num`.

Function

```
def score_candidates_map(alpha_num, alpha_den, known_size) -> Callable[[int], int]:
    def stability_map(d_in):
        if known_size:
            return T.inf_cast(d_in // 2).inf_mul(alpha_den)
        else:
            abs_dist_const: u64 = max(alpha_num, alpha_den - alpha_num)  #
            return T.exact_int_cast(d_in).alerting_mul(abs_dist_const)

    return stability_map
```

Postcondition

Theorem 1.1. Define `function`as follows:

$$\texttt{function}(x) = \text{score_candidates}_i(x, \texttt{candidates}, \texttt{alpha_num}, \texttt{alpha_den}, \texttt{size_limit})$$

The function calls `score_candidates` with fixed choices of `candidates`, `alpha_num`, `alpha_den` and `size_limit`.

If the input domain is the set of all vectors with non-null elements of type `TI`, input metric is either `SymmetricDistance` or `InsertDeleteDistance`, and the postcondition of `function`is satisfied, then `function`is stable.

This means that, for any two elements x, x' in the `input_domain`, where x, x' share the same length if `known_length` is true, and any pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_metric`, then when x, x' are `d_in`-close under `input_metric` and `stability_map`(`d_in`) $\leq$ `d_out`, `function`(x), `function`(x') are `d_out`-close under `output_metric`.

The sensitivity of this function differs depending on whether the size of the input vector is known. First, consider the case where the size is unknown.

[1] See new changes with `git diff f5bb719..db2ab1a rust/src/transformations/quantile_score_candidates/mod.rs`

Lemma 1.2. If $d_{Sym}(x, x') = 1$, then $d_\infty(\texttt{function}(x), \texttt{function}(x')) \leq \max(\alpha_{num}, \alpha_{den} - \alpha_{num})$.

Proof. Assume $d_{Sym}(x, x') = 1$.

$$
\begin{aligned}
& d_\infty(\texttt{function}(x)_i, \texttt{function}(x')_i) \\
&= \max_i |\texttt{function}(x)_i - \texttt{function}(x')_i| && \text{by definition of } d_\infty \\
&= \max_i |\text{abs_diff}(\alpha_{den} \cdot \min(\#(x < C_i), l), \alpha_{num} \cdot \min(|x| - \#(x = C_i), l)) && \text{by definition of } \texttt{function} \\
&\qquad \text{abs_diff}(\alpha_{den} \cdot \min(\#(x' < C_i), l), \alpha_{num} \cdot \min(|x'| - \#(x' = C_i), l))| \\
&= \alpha_{den} \cdot \max_i ||\min(\#(x < C_i), l) - \alpha \cdot \min(|x| - \#(x = C_i), l)| \\
&\qquad |\min(\#(x' < C_i), l) - \alpha \cdot \min(|x'| - \#(x' = C_i), l)|| \\
&\leq \alpha_{den} \cdot \max_i ||\#(x < C_i) - \alpha \cdot (|x| - \#(x = C_i))| \\
&\qquad |\#(x' < C_i) - \alpha \cdot (|x| - \#(x' = C_i))||
\end{aligned}
$$

Consider each of the three cases of adding or removing an element in x.

Case 1. Assume x' is equal to x, but with some $x_j < C_i$ added or removed.

$$
\begin{aligned}
&= \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - |(1-\alpha) \cdot (\#(x < C_i) \pm 1) - \alpha \cdot \#(x > C_i)|| \\
&\leq \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - (|(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| + |\pm(1-\alpha)|)| && \text{by triangle inequality} \\
&= \alpha_{den} \cdot \max_i |1 - \alpha| && \text{scores cancel} \\
&= \alpha_{den} - \alpha_{num} && \text{since } \alpha \leq 1
\end{aligned}
$$

Case 2. Assume x' is equal to x, but with some $x_j > C_i$ added or removed.

$$
\begin{aligned}
&= \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - |(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot (\#(x > C_i) \pm 1)|| \\
&\leq \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - (|(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| + |\pm\alpha|)| && \text{by triangle inequality} \\
&= \alpha_{den} \cdot \max_i |\alpha| && \text{scores cancel} \\
&= \alpha_{num} && \text{since } \alpha \geq 0
\end{aligned}
$$

Case 3. Assume x' is equal to x, but with some $x_j = C_i$ added or removed.

$$
\begin{aligned}
&= \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - |(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)|| \\
&= 0 && \text{no change in score}
\end{aligned}
$$

Take the union bound over all cases.

$$
\leq \max(\alpha_{num}, \alpha_{den} - \alpha_{num})
$$

□

Now consider the case where the dataset size is known.

Lemma 1.3. If $d_{CO}(x, x') \leq 1$, then $d_\infty(\texttt{function}(x), \texttt{function}(x')) \leq \alpha_{den}$.

Proof. Assume $d_{CO}(x, x') \leq 1$.

$$
\begin{aligned}
& d_\infty(\texttt{function}(x), \texttt{function}(x')) \\
&= \max_i |\texttt{function}(x)_i - \texttt{function}(x')_i| && \text{by definition of } d_\infty \\
&= \max_i |\text{abs_diff}(\alpha_{den} \cdot \min(\#(x < C_i), l), \alpha_{num} \cdot \min(|x| - \#(x = C_i), l)) && \text{by def. of } \texttt{function} \\
&\qquad - \text{abs_diff}(\alpha_{den} \cdot \min(\#(x' < C_i), l), \alpha_{num} \cdot \min(|x'| - \#(x' = C_i), l))| \\
&= \alpha_{den} \cdot \max_i ||\min(\#(x < C_i), l) - \alpha \cdot \min(|x| - \#(x = C_i), l)| \\
&\qquad - |\min(\#(x' < C_i), l) - \alpha \cdot \min(|x'| - \#(x' = C_i), l)|| \\
&= \alpha_{den} \cdot \max_i ||\#(x < C_i) - \alpha \cdot (|x| - \#(x = C_i))| \\
&\qquad - |\#(x' < C_i) - \alpha \cdot (|x| - \#(x' = C_i))||
\end{aligned}
$$

Consider each of the four cases of changing a row in x.

Case 1. Assume x' is equal to x, but with some $x_j < C_i$ replaced with $x'_j > C_i$.

$$
\begin{aligned}
&= \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - |(1-\alpha) \cdot (\#(x < C_i) - 1) - \alpha \cdot (\#(x > C_i) + 1)|| && \text{by definition of } \texttt{function} \\
&\leq \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - (|(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| + |1|)| && \text{by triangle inequality} \\
&= \alpha_{den} \cdot \max_i |1| && \text{scores cancel} \\
&= \alpha_{den}
\end{aligned}
$$

Case 2. Assume x' is equal to x, but with some $x_j > C_i$ replaced with $x'_j < C_i$.

$$= \alpha_{den}$$

by symmetry, follows from Case 1.

Case 3. Assume x' is equal to x, but with some $x_j \neq C_i$ replaced with C_i.

$$\leq \max(\alpha_{num}, \alpha_{den} - \alpha_{num})$$

equivalent to one removal (see `make_quantile_score_candidates`)

Case 4. Assume x' is equal to x, but with some $x_j = C_i$ replaced with $x'_j \neq C_i$.

$$\leq \max(\alpha_{num}, \alpha_{den} - \alpha_{num})$$

equivalent to one addition (see `make_quantile_score_candidates`)

Take the union bound over all cases.

$$d_\infty(x_i, x'_i) \leq \max(\alpha_{den}, \max(\alpha_{num}, \alpha_{den} - \alpha_{num})) = \alpha_{den}$$

since $\max(\alpha, 1-\alpha) \leq 1$

□

Proof of postcondition. Assume the input domain is the set of all vectors with non-null elements of type `TI`, input metric is either `SymmetricDistance` or `InsertDeleteDistance`, and the postcondition of `function`is satisfied.

First, consider the case where the size is unknown. Take any two members s, s' in the `input_domain` and any pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_metric`. Assume s, s' are `d_in`-close under `input_metric` and that `stability_map`(`d_in`) $\leq$ `d_out`.

$$\begin{aligned}
\texttt{d_out} &= \max_{s \sim s'} d_\infty(s, s') && \text{where } s = \texttt{function}(x) \\
&= \max_{s \sim s'} \max_i |s_i - s'_i| && \text{by definition of } \texttt{LInfDistance}\text{, without monotonicity} \\
&\leq \sum_j^{\texttt{d_in}} \max_{Z_j \sim Z_{j+1}} \max_i |s_{i,j} - s_{i,j+1}| && \text{by path property } d_{Sym}(Z_i, Z_{i+1}) = 1, x = Z_0 \text{ and } x' = Z_{\texttt{d_in}} \\
&\leq \sum_j^{\texttt{d_in}} \max(\alpha_{num}, \alpha_{den} - \alpha_{num}) && \text{by 1.2} \\
&\leq \texttt{d_in} \cdot \max(\alpha_{num}, \alpha_{den} - \alpha_{num})
\end{aligned}$$

This formula matches the stability map in the case where the dataset size is unknown.

Now, consider the case where the size is known. Take any two elements s, s' in the `input_domain` and any pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_metric`. Assume s, s' are `d_in`-close under `input_metric` and that `stability_map`(`d_in`) $\leq$ `d_out`.

$$
\begin{aligned}
\texttt{d_out} &= \max_{s \sim s'} d_\infty(s, s') \\
&= \max_{s \sim s'} \max_i |s_i - s'_i| && \text{by definition of } \texttt{LInfDistance}\text{, without monotonicity} \\
&\leq \sum_j^{\texttt{d_in}//2} \max_{Z_j \sim Z_{j+1}} \max_i |s_{i,j} - s_{i,j+1}| && \text{by path property } d_{CO}(Z_i, Z_{i+1}) = 1, x = Z_0 \text{ and } Z_{\texttt{d_in}} = x' \\
&\leq \sum_j^{\texttt{d_in}//2} \alpha_{den} && \text{by 1.3} \\
&\leq (\texttt{d_in}//2) \cdot \alpha_{den}
\end{aligned}
$$

This formula matches the stability map in the case where the dataset size is known.

It is shown that `function`(x), `function`(x') are `d_out`-close under `output_metric` for any choice of input arguments. □